

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Rounding, Estimation & Bounds

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ROUNDING, ESTIMATION & BOUNDS · CH1 · L9



ENG. ESLAM AHMED

The Map

This lesson collapses Rounding, Estimation & Bounds into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Round to the requested accuracy

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find lower and upper bounds

TRIGGER *"work out", "find", a missing value*

03 Choose bounds for largest value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Choose bounds for smallest value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Estimate the value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Find upper bound

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 37 website questions, 37 checked solutions, 3 local PDFs read.

01 Round to the requested accuracy

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rounding, Estimation & Bounds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$X = \text{area of rectangle} - \text{area of square}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2025 P2H Q22

The shaded area is

$$X = \text{area of rectangle} - \text{area of square}$$

Bounds

$$11.25 \leq AB < 11.75$$

Finish: $X = 90 \text{ cm}^2$ to a suitable degree of accuracy

FORMULA BANK

Half the rounding unit gives lower/upper bounds. Choose upper/lower to make the final expression largest or smallest.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

22 The diagram shows a square inside rectangle ABCD. A B D C Diagram NOT accurately drawn. The total area of the region shown shaded in the diagram is $X \text{ cm}^2$. $AB = 11.5 \text{ cm}$ correct to the nearest 0.5 cm . $BC = 9.2 \text{ cm}$ correct to 2 significant figures. Side of square = 4.1 cm correct to 2 significant figures. By considering bounds, work out the value of X to a suitable degree of accuracy. Show your working clearly. $X =$

YOUR SOLUTION

02 Find lower and upper bounds

TRIGGER *"work out", "find", a missing value*

BECOMES A Rounding, Estimation & Bounds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.25 \leq w < 4.35$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q11

Estimate the value

$$4.25 \leq w < 4.35$$

Find upper bound

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) 4.35 kg, (b) 4.25 kg, (c) 8.15

FORMULA BANK

Half the rounding unit gives lower/upper bounds. Choose upper/lower to make the final expression largest or smallest.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

11 The weight of a cat is 4.3 kg correct to 2 significant figures. (a) Write down the upper bound of the weight of the cat. kg (1) (b) Write down the lower bound of the weight of the cat. kg (1) G = e - f e = 17 correct to the nearest integer f = 9.4 correct to one decimal place (c) Work out the upper bound for the value of G. (2)

YOUR SOLUTION

03 Choose bounds for largest value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rounding, Estimation & Bounds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Half the rounding unit gives lower/upper bounds. Choose upper/lower to make the final expression largest or smallest.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Choose bounds for smallest value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rounding, Estimation & Bounds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Half the rounding unit gives lower/upper bounds. Choose upper/lower to make the final expression largest or smallest.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Estimate the value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rounding, Estimation & Bounds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Half the rounding unit gives lower/upper bounds. Choose upper/lower to make the final expression largest or smallest.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

16 $P = 2a \times c \times d - a = 58.4$ correct to 3 significant figures. $c = 20$ correct to 2 significant figures. $d = 3.6$ correct to 2 significant figures. Work out the upper bound for the value of P . Show your working clearly. Give your answer correct to 2 decimal places.

YOUR SOLUTION

06 Find upper bound

TRIGGER *"work out", "find", a missing value*

BECOMES A Rounding, Estimation & Bounds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Half the rounding unit gives lower/upper bounds. Choose upper/lower to make the final expression largest or smallest.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

16 $P = 2a \times c \times d - a = 58.4$ correct to 3 significant figures. $c = 20$ correct to 2 significant figures. $d = 3.6$ correct to 2 significant figures. Work out the upper bound for the value of P . Show your working clearly. Give your answer correct to 2 decimal places.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Round to the requested accuracy
"work out", "find", a missing value	Find lower and upper bounds
"work out", "show", "hence", a familiar exam phrase	Choose bounds for largest value
"work out", "show", "hence", a familiar exam phrase	Choose bounds for smallest value
"work out", "show", "hence", a familiar exam phrase	Estimate the value
"work out", "find", a missing value	Find upper bound
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.